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Generalized coupled thermoelasticity of functionally graded cylindrical shells

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SUMMARY

In this paper, the response of a circular cylindrical thin shell made of the functionally graded material based on the generalized theory of thermoelasticity is obtained. The governing equations of the generalized theory of thermoelasticity and the energy equations are simultaneously solved for a functionally graded axisymmetric cylindrical shell subjected to thermal shock load. Thermoelasticity with second sound effect in cylindrical shells based on the Lord–Shulman model is compared with the Green–Lindsay model. A second-order shear deformation shell theory, that accounts for the transverse shear strains and rotations, is considered. Including the thermo-mechanical coupling and rotary inertia, a Galerkin finite element formulation in space domain and the Laplace transform in time domain is used to formulate the problem. The inverse Laplace transform is obtained using a numerical algorithm. The shell is graded through the thickness assuming a volume fraction of metal and ceramic, using a power law distribution. The effects of temperature field for linear and non-linear distributions across the shell thickness are examined. The results are validated with the known data in the literature. Copyright © 2006 John Wiley & Sons, Ltd.

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1. INTRODUCTION

The classical theory of heat conduction assumes that if a media conducting heat is subjected to a thermal signal, the effect of the signal will be felt at distances infinitely far from its source, instantaneously. However, empirical studies do not support this assumption. Lord–Shulman [1]

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and Green–Lindsay [2] presented two different theories to eliminate the paradox of infinite heat propagation speed. The associate theories are referred to the 'generalized' theories of thermoelasticity with second sound effect. The 'generalized' coupled thermoelasticity theory means the coupled thermoelasticity with relaxation time(s).

The main paradox starts from this point that, the conventional heat conduction equation is parabolic-type partial differential equation, and allows an infinite speed for thermal signals. Converting the heat conduction equation to a hyperbolic-type equation, is the aim of generalized thermoelasticity. Lord–Shulman formulated a model by using the non-Fourier thermal wave model. This was done by introducing a flux-rate term into the heat conduction equation.

Green–Lindsay used an entropy production inequality proposed by Green and Laws [3] and so a new energy function by including the temperature-rate among the constitutive variables. In 1980, Dhaliwal and Sherief [4] obtained the equations of generalized thermoelasticity for the anisotropic case. Green and Naghdi [5] presented the thermoelasticity without energy dissipation in 1991. Their constitutive equations is based on a reduced equation of balance of energy. They formulated the constitutive equations without using the entropy production inequality. In 1995, Tzou [6] introduced a new phase-lag of the temperature gradient in addition to the phase-lag of the heat flux named the dual-phase-lag model. Erbay and Suhubi [7] studied the free longitudinal waves in an infinite circular cylinder. Sherief and Hamza [8] considered the problem of a thick infinite plate whose stress-free faces are suddenly raised to a radially varying temperature. Khoma [9] derived the governing equations applicable to anisotropic shells of varying thickness. Eslami *et al*. [10] studied the coupled thermoelasticity of long thin cylindrical shells. They used a linear temperature distribution across the thickness of the shell, employing the classical coupled theory of thermoelasticity. Eslami *et al*. [11] studied the coupled thermoelasticity of shells of revolution. They employed the Lord–Shulman coupled thermoelasticity model with the linear temperature distribution across the shell thickness, and studied the effect of normal stress and coupling. Attempts on functionally graded materials are restricted to the conventional thermoelastic theory. Shiari *et al*. [12] solved the thermomechanical shocks in a multilayer orthotropic composite cylindrical shells. Bahtui and Eslami [13] presented the coupled thermoelastic response of a functionally graded circular cylindrical shell. They considered the linear classical coupled thermoelastic theory, and studied the effect of functionally graded properties on the coupled stress fields. They also examined the convergence and accuracy of two different types of elements in the finite element solution. Reddy and Chin [14] studied a thermomechanical analysis of functionally graded thick cylinders and thin plates. They assumed temperature-dependent properties in order to obtain more realistic results.

The present paper deals with the generalized coupled thermoelasticity problem of a functionally graded cylindrical shell under impulsive thermal shock load. The material properties are graded across the thickness direction according to a volume fraction power law distribution. The governing equations are based on the second-order shear deformation shell theory, including the rotary inertia term. The thermoelasticity with thermal relaxation (Lord–Shulman model) and the temperature-rate-dependent thermoelasticity (Green–Lindsay model) are considered, and the Galerkin finite element formulation is used. The governing equations are transformed into the Laplace domain, where the Galerkin finite element method is used to obtain the solution in the space domain. The inversion of the Laplace transform into the physical time domain is obtained numerically. A second-order polynomial for the temperature distribution across the shell thickness is considered, and the results are compared with the linear temperature distribution. The numerical results for the temperature, displacement, and the thermal stresses are obtained. Since the matrices in the

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numerical solution are of ill-condition type, a method is introduced to normalize ill-condition matrices. The results are validated with the known data in the literature [10, 11, 13].

2. ANALYSIS

In order to calculate the temperature and thermal stress distributions along the shell thickness, the material properties of the functionally graded shell, such as Young's modulus, thermal expansion coefficient, thermal conduction coefficient , specific heat conduction, and density must be described across the shell thickness. Since the shell material is assumed to be graded across the thickness direction, the shell material properties are assumed to be described with the power law as

$$
\begin{aligned}
E(z) &= E_{\mathrm{c}} + (E_{\mathrm{m}} - E_{\mathrm{c}})\left(\frac{z}{h}+\frac{1}{2}\right)^{m_1} \\
\alpha(z) &= \alpha_{\mathrm{c}} + (\alpha_{\mathrm{m}} - \alpha_{\mathrm{c}})\left(\frac{z}{h}+\frac{1}{2}\right)^{m_2} \\
k(z) &= k_{\mathrm{c}} + (k_{\mathrm{m}} - k_{\mathrm{c}})\left(\frac{z}{h}+\frac{1}{2}\right)^{m_3} \\
\rho(z) &= \rho_{\mathrm{c}} + (\rho_{\mathrm{m}} - \rho_{\mathrm{c}})\left(\frac{z}{h}+\frac{1}{2}\right)^{m_4} \\
c(z) &= c_{\mathrm{c}} + (c_{\mathrm{m}} - c_{\mathrm{c}})\left(\frac{z}{h}+\frac{1}{2}\right)^{m_5}
\end{aligned}
\tag{1}
$$

where the volume fractions indices m_i $(i=1,\ldots,5)$ represent the material variation profiles through the shell thickness, and indices c and m are abbreviations of ceramic and metal, respectively. It is assumed that the Poisson's ratio v is constant across the shell thickness.

2.1. Strain–displacement relations

Consider a functionally graded circular cylindrical shell. The cylindrical coordinates (x, θ, z) are considered along the axial, circumferential, and radial directions, respectively. In order to include the shear deformations $(\gamma_{xz}, \gamma_{\theta z} \neq 0)$, we may assume that the displacement components, based on the first-order approximation, are represented as

$$
\begin{aligned}
u(x,\theta,z) &= u_0(x,\theta) + z\psi_x(x,\theta,0) \\
v(x,\theta,z) &= v_0(x,\theta) + z\psi_\theta(x,\theta,0) \\
w(x,\theta,z) &= w_0(x,\theta)
\end{aligned}
\tag{2}
$$

where u_0, v_0, and w_0 represent the components of the displacement vector of a point on the middle plane, and ψ_x and ψ_θ represent the rotations of tangents to the middle plane along the x and θ

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axes, respectively. The strain distributions across the shell thickness are given as [15]

$$
\begin{aligned}
\varepsilon_{xx} &= \varepsilon_{xx}^{0} + z k_x \\
\varepsilon_{\theta\theta} &= \frac{1}{1+z/R}(\varepsilon_{\theta\theta}^{0} + z k_\theta) \\
\gamma_{x\theta} &= \gamma_{x\theta}^{0} + z k_{x\theta} \\
\gamma_{xz} &= \gamma_{xz}^{0} \\
\gamma_{\theta z} &= \frac{1}{1+z/R}\gamma_{\theta z}^{0}
\end{aligned}
\tag{3}
$$

where the strains and curvatures of the middle plane are defined as [15]

$$
\begin{aligned}
\varepsilon_{xx}^{0} &= u_{0,x} \\
\varepsilon_{\theta\theta}^{0} &= \frac{1}{R}(v_{0,\theta} + w_0) \\
\gamma_{x\theta}^{0} &= v_{0,x} + \frac{1}{R+z}u_{0,\theta} \\
\gamma_{xz}^{0} &= w_{0,x} + \psi_x \\
\gamma_{\theta z}^{0} &= \frac{1}{R}(w_{0,\theta} - v_0) + \psi_\theta \\
k_x &= \psi_{x,x} \\
k_\theta &= \frac{1}{R}\psi_{\theta,\theta} \\
k_{x\theta} &= \psi_{\theta,x} + \frac{1}{1+z/R}\psi_{x,\theta}
\end{aligned}
\tag{4}
$$

A comma denotes partial differentiation with respect to the space variables.

The stress–strain relations for a functionally graded shell based on the assumed displacement model, including the shear deformations and considering the generalized theory of thermoelasticity using the Green–Lindsay model are

$$
\begin{Bmatrix} \sigma_{xx} \\ \sigma_{\theta\theta} \\ \tau_{x\theta} \\ \tau_{xz} \\ \tau_{\theta z} \end{Bmatrix} = \frac{E(z)}{1-v^2}\begin{bmatrix} 1 & v & 0 & 0 & 0 \\ v & 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1-v}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1-v}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{1-v}{2} \end{bmatrix}\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{\theta\theta} \\ \gamma_{x\theta} \\ \gamma_{xz} \\ \gamma_{\theta z} \end{Bmatrix} - \frac{E(z)\alpha(z)}{1-v^2}\begin{Bmatrix} \Delta T + t_1\Delta\dot{T} \\ \Delta T + t_1\Delta\dot{T} \\ 0 \\ 0 \\ 0 \end{Bmatrix}
\tag{5}
$$

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where $\Delta T(x, \theta, z, t)$ is the temperature change, and t_1 is the relaxation time used in the Green–Lindsay model. The force and moment resultants from the second-order shell theory are

$$\begin{Bmatrix} N_{xx} \\ N_{x\theta} \\ Q_{xx} \end{Bmatrix} = \int_z \begin{Bmatrix} \sigma_{xx} \\ \tau_{x\theta} \\ \tau_{xz} \end{Bmatrix} \left(1 + \frac{z}{R}\right) \mathrm{d}z$$

$$\begin{Bmatrix} N_{\theta\theta} \\ N_{\theta x} \\ Q_{\theta\theta} \end{Bmatrix} = \int_z \begin{Bmatrix} \sigma_{\theta\theta} \\ \tau_{\theta x} \\ \tau_{\theta z} \end{Bmatrix} \mathrm{d}z \tag{6}$$

$$\begin{Bmatrix} M_{xx} \\ M_{x\theta} \end{Bmatrix} = \int_z \begin{Bmatrix} \sigma_{xx} \\ \tau_{x\theta} \end{Bmatrix} \left(1 + \frac{z}{R}\right) z \, \mathrm{d}z$$

$$\begin{Bmatrix} M_{\theta\theta} \\ M_{\theta x} \end{Bmatrix} = \int_z \begin{Bmatrix} \sigma_{\theta\theta} \\ \tau_{\theta x} \end{Bmatrix} z \, \mathrm{d}z$$

2.2. Equations of motion

Love's equations of motion in cylindrical coordinates, including the shear deformations, are [16]

$$\begin{aligned} N_{xx,x} + \frac{1}{R} N_{\theta x,\theta} &= I_1 \ddot{u}_0 \\ N_{x\theta,x} + \frac{1}{R} N_{\theta\theta,\theta} + \frac{1}{R} Q_{\theta\theta} &= I_1 \ddot{v}_0 \\ Q_{xx,x} + \frac{1}{R} Q_{\theta\theta,\theta} - \frac{1}{R} N_{\theta\theta} &= p_z(t) + I_1 \ddot{w}_0 \\ M_{xx,x} + \frac{1}{R} M_{\theta x,\theta} - Q_x &= I_2 \ddot{\psi}_x \\ M_{x\theta,x} + \frac{1}{R} M_{\theta,\theta} - Q_\theta &= I_2 \ddot{\psi}_\theta \end{aligned} \tag{7}$$

A dot denotes partial differentiation with respect to time and

$$\begin{aligned} I_1 &= \int_z \rho(z) \, \mathrm{d}z \\ I_2 &= \int_z \rho(z) z^2 \left(1 + \frac{z}{R}\right) \mathrm{d}z \end{aligned} \tag{8}$$

In terms of the applied forces to the inside and outside surfaces of the cylindrical shell, $p_z(t)$ is

$$p_z(t) = p_z^+(t) \left(1 + \frac{h}{2R}\right) + p_z^-(t) \left(1 - \frac{h}{2R}\right) \tag{9}$$

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where p_z^+ and p_z^- are the applied external and internal pressures, respectively. The sign of $p_z(t)$ is assumed positive toward the centre.

Since the thermal and mechanical applied loads are assumed to be axisymmetric, then

$$\frac{\partial}{\partial\theta}=0$$

leaving three equations of motion as follows to be simultaneously solved with the energy equation:

$$\begin{aligned} N_{xx,x}+\frac{1}{R}N_{\theta x,\theta}&=I_1\ddot{u}_0\\ Q_{xx,x}+\frac{1}{R}Q_{\theta\theta,\theta}-\frac{1}{R}N_{\theta\theta}&=p_z(t)+I_1\ddot{w}_0\\ M_{xx,x}+\frac{1}{R}M_{\theta x,\theta}-Q_x&=I_2\ddot{\psi}_x \end{aligned} \tag{10}$$

Temperature distribution across the shell thickness may be assumed to be linear (for very thin shells), or have a quadratic form. The effect of these two temperature distributions are compared further. For the linear approximation, the temperature distribution is assumed as

$$\Delta T_1(x,\theta,z,t)=T_0(x,\theta,t)+zT_1(x,\theta,t) \tag{11}$$

where T_0 and T_1 are both unknowns. For the quadratic distribution we assume a temperature distribution as

$$\Delta T_2(x,\theta,z,t)=T_0(x,\theta,t)+zT_1(x,\theta,t)+\tfrac{1}{2}z^2T_2(x,\theta,t) \tag{12}$$

where T_0, T_1, and T_2 are some unknown functions and must be obtain through the coupled system of equations.

Substituting Equations (1)–(6), (8)–(9), and (12) into Equations (10), yields three equations of motion as

$$\begin{aligned} &\left[\left(\Omega_1\Pi_{11}^1+\Omega_2\Pi_{11}^2\frac{\partial^2}{\partial x^2}-I_1\right)\frac{\partial^2}{\partial t^2}\right]u_0+\left[\frac{v}{R}(\Omega_1\Pi_{01}^1+\Omega_2\Pi_{01}^2)\frac{\partial}{\partial x}\right]w_0\\ &\quad+\left[(\Omega_1\Pi_{12}^1+\Omega_2\Pi_{12}^2)\frac{\partial^2}{\partial x^2}\right]\psi_x-\left[(\Omega_3\Pi_{11}^1+\Omega_4\Pi_{11}^2+\Omega_5\Pi_{11}^3)\left(\frac{\partial}{\partial x}+t_1\frac{\partial^2}{\partial x\partial t}\right)\right]T_0\\ &\quad-\left[(\Omega_3\Pi_{12}^1+\Omega_4\Pi_{12}^2+\Omega_5\Pi_{12}^3)\left(\frac{\partial}{\partial x}+t_1\frac{\partial^2}{\partial x\partial t}\right)\right]T_1\\ &\quad-\left[(\Omega_3\Pi_{13}^1+\Omega_4\Pi_{13}^2+\Omega_5\Pi_{13}^3)\left(\frac{\partial}{\partial x}+t_1\frac{\partial^2}{\partial x\partial t}\right)\right]T_2=0 \end{aligned}$$

$$-\left[\frac{v}{R}(\Omega_1\Pi_{01}^1+\Omega_2\Pi_{01}^2)\frac{\partial}{\partial x}\right]u_0+\left[(\Omega_6\Pi_{11}^1+\Omega_7\Pi_{11}^2)\frac{\partial^2}{\partial x^2}-\frac{1}{R^2}(\Omega_1\Pi_{21}^1+\Omega_2\Pi_{21}^2)-I_1\frac{\partial^2}{\partial t^2}\right]w_0$$

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$$+\left[\left(\Omega_6\Pi_{11}^1+\Omega_7\Pi_{11}^2-\Omega_2\Pi_{02}^2\frac{v}{R}\right)\frac{\partial}{\partial x}\right]\psi_x+\left[\frac{1}{R}(\Omega_3\Pi_{01}^1+\Omega_4\Pi_{01}^2+\Omega_5\Pi_{01}^3)\left(1+t_1\frac{\partial}{\partial t}\right)\right]T_0$$

$$+\left[\frac{1}{R}(\Omega_4\Pi_{02}^2+\Omega_5\Pi_{02}^3)\left(1+t_1\frac{\partial}{\partial t}\right)\right]T_1+\left[\frac{1}{R}(\Omega_3\Pi_{03}^1+\Omega_4\Pi_{03}^2+\Omega_5\Pi_{03}^3)\left(1+t_1\frac{\partial}{\partial t}\right)\right]T_2=p_z(t)$$

$$\left[(\Omega_1\Pi_{12}^1+\Omega_2\Pi_{12}^2)\frac{\partial^2}{\partial x^2}\right]u_0+\left[\left(\Omega_2\Pi_{02}^2\frac{v}{R}-\Omega_6\Pi_{11}^1-\Omega_7\Pi_{11}^2\right)\frac{\partial}{\partial x}\right]w_0$$

$$+\left[(\Omega_1\Pi_{13}^1+\Omega^2\Pi_{13}^2)\frac{\partial^2}{\partial x^2}-\Omega_6\Pi_{11}^1-\Omega_7\Pi_{11}^2-I_2\frac{\partial^2}{\partial t^2}\right]\psi_x$$

$$-\left[(\Omega_3\Pi_{12}^1+\Omega_4\Pi_{12}^2+\Omega_5\Pi_{12}^3)\left(\frac{\partial}{\partial x}+t_1\frac{\partial^2}{\partial x\partial t}\right)\right]T_0$$

$$-\left[(\Omega_3\Pi_{13}^1+\Omega_4\Pi_{13}^2+\Omega_5\Pi_{13}^3)\left(\frac{\partial}{\partial x}+t_1\frac{\partial^2}{\partial x\partial t}\right)\right]T_1$$

$$-\left[(\Omega_3\Pi_{14}^1+\Omega_4\Pi_{14}^2+\Omega_5\Pi_{14}^3)\left(\frac{\partial}{\partial x}+t_1\frac{\partial^2}{\partial x\partial t}\right)\right]T_2=0 \tag{13}$$

where constants Ω_i $(i=1,\ldots,7)$ and Π_{jk}^i $(i=1,2,3;\ j=0,1,2;\ k=0,\ldots,4)$ are given in the appendix.

2.3. *Energy equation*

The first law of thermodynamics for heat conduction equation in the assumed cylindrical functionally graded shell, when the temperature distribution is assumed to be a function of time and thickness direction, is

$$\rho(z)c(z)(t_0+t_2)\Delta\ddot{T}+\rho(z)c(z)\Delta\dot{T}+\beta_{ij}(z)T_a\left(\frac{\partial}{\partial t}+t_0\frac{\partial^2}{\partial t^2}\right)u_{i,j}-(k(z)\Delta T_{,j})_{,i}=0 \tag{14}$$

where $\beta_{ij}(z)$ varies through the shell thickness, t_0, and t_2 are relaxation times according to the Lord–Shulman and Green–Lindsay models, respectively. The heat conduction equation (14) may be written in expanded form for the assumed conditions. Using the definition for the material properties given by Equation (1), Equation (14) becomes

$$\begin{aligned}\mathrm{Re}(T)=&\ \rho(z)c(z)(t_0+t_2)\Delta\ddot{T}+\rho(z)c(z)\Delta\dot{T}\\&+\alpha(z)E(z)\frac{T_a}{1-v}\times\left(\frac{\partial}{\partial t}+t_0\frac{\partial^2}{\partial t^2}\right)\left[\frac{\partial u}{\partial x}+\frac{w}{R+z}\right]-k(z)\frac{\partial^2 T}{\partial x^2}\\&-\frac{k(z)}{R+z}\frac{\partial T}{\partial z}-\frac{\partial k(z)}{\partial z}\frac{\partial T}{\partial z}-k(z)\frac{\partial^2 T}{\partial z^2}=0\end{aligned} \tag{15}$$

The assumed temperature distribution from Equation (12) is substituted into Equation (15) and the resulting residue $\mathrm{Re}(T)$ is made orthogonal with respect to 1, z, and z^2. This yields three independent equations for T_0, T_1, and T_2, as

$$\begin{aligned}&\int_z \mathrm{Re}(T)\times \mathrm{d}z=0\\&\int_z \mathrm{Re}(T)\times z\,\mathrm{d}z=0\\&\int_z \mathrm{Re}(T)\times z^2\,\mathrm{d}z=0\end{aligned}\tag{16}$$

These three equations are related to T_0, T_1, and T_2, respectively, and are the final three equations required for the solution steps. A heat flux q_i is taken into consideration to apply the temperature boundary condition for the convection purpose as

$$q_i=h_i(T-T_\infty),\quad i=\text{in, out}\tag{17}$$

Substituting Equations (15) and (17) into Equations (16), yields three energy equations as

$$\begin{aligned}&\left[-T_a\Gamma_{10}^1\left(\frac{\partial^2}{\partial x\partial t}+t_0\frac{\partial^3}{\partial x\partial t^2}\right)\right]u_0-\left[\frac{T_a}{R}\Gamma_{12}^1\left(\frac{\partial}{\partial t}+t_0\frac{\partial^2}{\partial t^2}\right)\right]w_0-\left[T_a\Gamma_{20}^1\left(\frac{\partial^2}{\partial x\partial t}+t_0\frac{\partial^3}{\partial x\partial t^2}\right)\right]\psi_x\\&\quad+\left[\Gamma_{10}^2\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{10}^3\frac{\partial^2}{\partial x^2}+(h_{\text{in}}+h_{\text{out}})\right]T_0\\&\quad+\left[\Gamma_{20}^2\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{20}^3\frac{\partial^2}{\partial x^2}-\frac{\Gamma_{12}^3}{R}+\frac{h}{2}(h_{\text{out}}-h_{\text{in}})\right]T_1\\&\quad+\left[\Gamma_{30}^2\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{30}^3\frac{\partial^2}{\partial x^2}-\frac{2}{R}\Gamma_{22}^3+\frac{h^2}{4}(h_{\text{out}}+h_{\text{in}})\right]T_2\\&\quad=h_{\text{in}}T_{\text{in}}(t)+h_{\text{out}}T_{\text{out}}(t)-T_a(h_{\text{in}}+h_{\text{out}})\\&\left[-T_a\Gamma_{20}^1\left(\frac{\partial^2}{\partial x\partial t}+t_0\frac{\partial^3}{\partial x\partial t^2}\right)\right]u_0-\left[\frac{T_a}{R}\Gamma_{22}^1\left(\frac{\partial}{\partial t}+t_0\frac{\partial^2}{\partial t^2}\right)\right]w_0-\left[T_a\Gamma_{30}^1\left(\frac{\partial^2}{\partial x\partial t}+t_0\frac{\partial^3}{\partial x\partial t^2}\right)\right]\psi_x\\&\quad+\left[\Gamma_{20}^2\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{20}^3\frac{\partial^2}{\partial x^2}+\frac{h}{2}(h_{\text{out}}-h_{\text{in}})\right]T_0\\&\quad+\left[\Gamma_{30}^2\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{30}^3\frac{\partial^2}{\partial x^2}-\frac{\Gamma_{22}^3}{R}+\frac{h^2}{4}(h_{\text{out}}+h_{\text{in}})+\Gamma_{10}^3\right]T_1\end{aligned}$$

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$$+\left[\Gamma_{40}^{2}\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{40}^{3}\frac{\partial^2}{\partial x^2}-\frac{2}{R}\Gamma_{32}^{3}+\frac{h^3}{8}(h_{\text{out}}-h_{\text{in}})+2\Gamma_{20}^{3}\right]T_2$$

$$=\frac{h}{2}(h_{\text{out}}T_{\text{out}}(t)-h_{\text{in}}T_{\text{in}}(t)+T_a(h_{\text{in}}-h_{\text{out}}))$$

$$\left[-T_a\Gamma_{30}^{1}\left(\frac{\partial^2}{\partial x\partial t}+t_0\frac{\partial^3}{\partial x\partial t^2}\right)\right]u_0-\left[\frac{T_a}{R}\Gamma_{32}^{1}\left(\frac{\partial}{\partial t}+t_0\frac{\partial^2}{\partial t^2}\right)\right]w_0-\left[T_a\Gamma_{40}^{1}\left(\frac{\partial^2}{\partial x\partial t}+t_0\frac{\partial^3}{\partial x\partial t^2}\right)\right]\psi_x$$

$$+\left[\Gamma_{30}^{2}\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{30}^{3}\frac{\partial^2}{\partial x^2}+\frac{h^2}{4}(h_{\text{out}}+h_{\text{in}})\right]T_0$$

$$+\left[\Gamma_{40}^{2}\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{40}^{3}\frac{\partial^2}{\partial x^2}-\frac{\Gamma_{32}^{3}}{R}+\frac{h^3}{8}(h_{\text{out}}-h_{\text{in}})+2\Gamma_{20}^{3}\right]T_1$$

$$+\left[\Gamma_{50}^{2}\left(\frac{\partial}{\partial t}+(t_0+t_2)\frac{\partial^2}{\partial t^2}\right)-\Gamma_{50}^{3}\frac{\partial^2}{\partial x^2}-\frac{2}{R}\Gamma_{42}^{3}+\frac{h^4}{16}(h_{\text{out}}+h_{\text{in}})+4\Gamma_{30}^{3}\right]T_2$$

$$=\frac{h^2}{4}(h_{\text{out}}T_{\text{out}}(t)+h_{\text{in}}T_{\text{in}}(t)-T_a(h_{\text{in}}+h_{\text{out}})) \tag{18}$$

where constants Γ_{jk}^{i} ($i=1,2,3$; $j=1,\dots,5$; $k=0,1,2$) are given in the appendix. Three coupled thermoelasticity models are obtained choosing t_0, t_1, and t_2 as

1. Classical model, $t_0=t_1=t_2=0$.
2. Lord–Shulman model, $t_0\neq 0$ and $t_1=t_2=0$.
3. Green–Lindsay model, $t_0=0$ and $t_1, t_2\neq 0$.

In the numerical examples, these three models are compared.

The results are reduced to the shell with isotropic material using the classical model of thermoelasticity and validated with the known data in the literature [10]. Also, the results are reduced to the shell with isotropic material using the Lord–Shulman generalized model of thermoelasticity and validated with the known data in the literature [11]. Finally, the results are reduced to the classical model of thermoelasticity and validated with the known data in the literature [13].

3. NUMERICAL SOLUTION

The simultaneous solution of the axisymmetric equations of motion and the energy equation provide a system of equations to be solved to obtain the six unknown functions u_0, ψ_x, w_0, T_0, T_1, and T_2 as functions of time for the generalized thermoelasticity problem of functionally graded cylindrical shell.

The Laplace transform technique is used to transform the governing equations into the space domain, where Galerkin finite element method is employed to obtain the solution in the space domain. Inverse Laplace transform is performed, numerically, to obtain the final solution in the physical time domain. This technique allows to analyse and study the behaviour of the system in very short time intervals. The standard time marching methods may fail to provide precise solution in time domain, when time increments are selected to be very small. The method of inverse Laplace transform proposed by Durbin [17] is used to transform the finite element results into the real time domain. The Durbins modified method [17] has three errors: truncation error, roundoff error, and the constant error bound (Durbin named it ERROR3 in his paper). The truncation and roundoff errors do exist in any type of numerical analysis. How to reduce them is related to the choice of NSUM (terms of Fourier series) and the ϖ value (note: $s=\varpi+\mathrm{i}w$). Durbin found that $\varpi * T_t = 5$–10 (T_t is total dimensionless time period) gave good results for NSUM ranging from 50 to 5000. The higher choice of $\varpi * T_t$ and NSUM values, the more accurate results will obtain. Numerical solutions in this paper are obtain using $\varpi * T_t = 7.5$, NSUM$=4000$ (higher values of NSUM takes very longer computational time), and $\varpi = 7.5/T_t$.

In order to solve the governing equations in the spatial domain, the Galerkin finite element technique is used. Applying the Galerkin technique to the system of six Laplace transferred equations and employing the proper weak formulations, yields

$$[K(s)]^{(e)} \times \{X\}^{(e)} = \{F(s)\}^{(e)} \tag{19}$$

where $[K(s)]$ is the global stiffness matrix, $\{F(s)\}$ is the force matrix, and $\{X\}$ is the unknown matrix in the form

$$\{X\}^{(e)^{\mathrm{T}}} = \langle u_0 \ \psi_x \ w_0 \ T_0 \ T_1 \ T_2 \rangle^{(e)}$$

However, since the time domain is transformed into the s-domain (Laplace), the mass, capacitance, and stiffness element matrices are compressed into only one matrix, called the global stiffness matrix $[K(s)]^{(e)}$.

The governing equations are changed into dimensionless form through the following formulas:

$$\begin{aligned}
&x = \bar{x} Y_1, \quad t = \frac{\bar{t} Y_3}{\lambda_7} \\
&t_0 = \frac{\bar{t}_0 Y_3}{\lambda_7}, \quad t_1 = \frac{\bar{t}_1 Y_3}{\lambda_7}, \quad t_2 = \frac{\bar{t}_2 Y_3}{\lambda_7} \\
&T_0 = \frac{\bar{T}_0 Y_4}{\lambda_8}, \quad T_1 = \frac{\bar{T}_1 Y_4}{\lambda_8 Y_1}, \quad T_2 = \frac{\bar{T}_2 Y_4}{\lambda_8 Y_1^2} \\
&u_0 = \frac{\bar{u}_0 Y_1 Y_5}{\lambda_1 Y_6}, \quad w_0 = \frac{\bar{w}_0 Y_1 Y_5}{\lambda_1 Y_6} \\
&\psi_x = \frac{\bar{\psi}_x Y_5}{\lambda_3 Y_6} \\
&I_1 = \bar{I}_1 Y_1 \rho_{\mathrm{a}}, \quad I_2 = \bar{I}_2 Y_1^3 \rho_{\mathrm{a}} \\
&q_z = \bar{q} Y_7, \quad \sigma_{ij} = \bar{\sigma}_{ij} Y_5 \\
&N_{ii} = \frac{\bar{N}_{ii} Y_7}{Y_1}, \quad M_{ij} = \bar{M}_{ij} Y_7
\end{aligned} \tag{20}$$

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where

$$
\begin{aligned}
&Y_1 = \frac{L}{N}\lambda_5, \quad Y_2 = \frac{Y_2}{Y_1\lambda_4} \\
&Y_3 = T_a\lambda_6, \quad Y_4 = \frac{E_a\alpha_a Y_3}{1-v} \\
&Y_5 = \lambda_a + \mu_a\lambda_2, \quad Y_6 = p_{\max}
\end{aligned}
\tag{21}
$$

Here, λ_i $(i=1,2,\ldots,8)$ are some arbitrary constants which are selected with trial and errors. These constants help matrix normalization in the finite element method, and their role is to make the mass, capacitance, and stiffness matrices of the same order of magnitudes. As mentioned, in the case of s-domain (Laplace), the global stiffness matrix is substituted by other finite element matrices. Since the problem is of shock-type, the global stiffness matrix becomes ill-condition. Therefore, λ's are selected in such a way to normalize the global stiffness matrix.

4. RESULTS AND DISCUSSION

Consider a clamped functionally graded cylindrical shell under the inside impulsive thermal shock. The length of shell is assumed to be 500 mm, diameter 200 mm, and wall thickness of 10 mm. The ratio R/h is thus 10. The functionally graded shell is assumed to be made of combination of metal (Ti–6Al–4V) and ceramic (ZrO_2), at the initial temperature 298.15°K, with the material properties give in Reference [18] and shown in Table I. The shell is ceramic-rich at the inside and metal-rich at the outside surfaces, respectively. Due to the axial symmetry, one-half of the cylindrical shell is modelled along the mid-surface. Temperature field across the shell thickness is assumed to be of linear type. Thermal shock is of impulsive type and is applied to the inside surface. The equation of thermal shock is

$$T(t) = 2201.85 \times aT\mathrm{e}^{-13100bt} + 298.15^\circ\mathrm{K} \tag{22}$$

where a and b are selected such that the impulsive shock period becomes 10^{-6} s. Figure 1 shows the behaviour of thermal shock load. This equation generates a convective heat flux equal to $2201.85 \times h_{\mathrm{in}}\,\mathrm{W/m^2}$ in 0.5 µs. The thermal shock's period is therefore 1 µs, and figures are drawn for 2.2 µs. The coefficient of thermal convection of the ceramic-rich inside surface is ($h_{\mathrm{in}} = 10\,000\,\mathrm{W/m^2\,K}$) and that of the metal-rich outside surface is ($h_{\mathrm{out}} = 200\,\mathrm{W/m^2\,K}$).

Table I. Material properties of titanium and zirconia.

Metal: Ti–6Al–4V	Ceramic: ZrO_2
$E = 66.2$ (GPa)	$E = 117.0$ (GPa)
$v = 0.321$	$v = 0.333$
$\alpha = 10.3 \times 10^{-6}$ (1/K)	$\alpha = 7.11 \times 10^{-6}$ (1/K)
$\rho = 4.41 \times 10^3$ (kg/m^3)	$\rho = 5.6 \times 10^3$ (kg/m^3)
$k = 18.1$ (W/mK)	$k = 2.036$ (W/mK)
$c = 808.3$ (J/kg K)	$c = 615.6$ (J/kg K)

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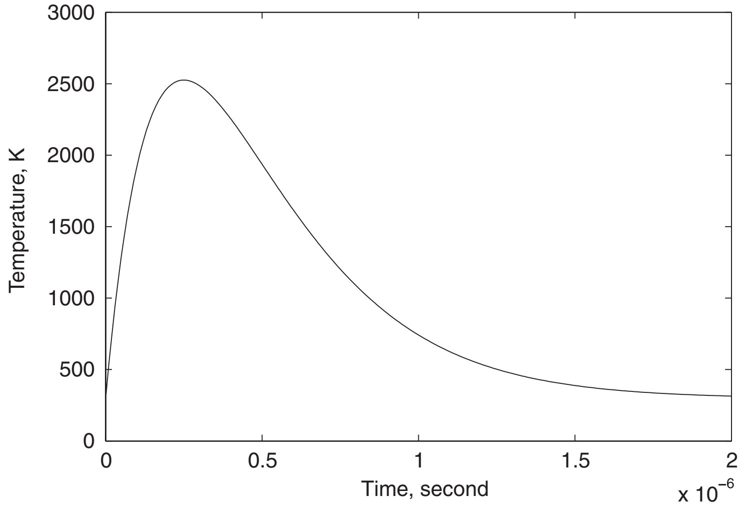

Figure 1. Impulsive thermal shock load.

The thermal boundary conditions at the ends of the shell are assumed insulated. The functionally graded shell is studied for the power law index of $m_i = 3$ $(i = 1, \ldots, 5)$.

In this example, the classical and two models of generalized coupled theories of thermoelasticity are compared. The Lord–Shulman thermal relaxation time t_0 of FGM is assumed as 1.5×10^{-6}. The Green–Lindsay thermal relaxation times t_1 and t_2 are taken to satisfy $t_1/t_2 = 2$ and $0.25 < t_2^{-0.5} < 1$. Solution convergence is determined with proper selection of λ_i $(i = 1, \ldots, 8)$. Proper selection of these constants will effect solution duration time and solution convergence. These constants help to normalize ill-condition matrices in the numerical solutions. In these example, the global stiffness matrix is attempted to be normalized. This procedure is done by trail and error method which leads to these λ's: $\lambda_1 = 10$, $\lambda_5 = 10$ and $\lambda_7 = 10^6$ (other λ parameters equal 1). The abbreviation of 'Classical', 'Lord–Shulman' and 'Green–Linsday' generalized models of thermoelasticity are taken as 'CL', 'LS' and 'GL', respectively.

The Durbins modified numerical inverse Laplace transform method [17] is used in this example and has two advantages: First, the error bound (ERROR3 in Durbin's paper) on the inverse $f(t)$ becomes independent of t (time variable), instead of being exponential in t; second, and consequently, the trigonometric series obtained for $f(t)$ in terms of $F(s)$ is valid on the whole period of the series. As it is proved in Durbin's paper, this error bound can be set arbitrarily small, and it is always possible to get good results, even in rather difficult cases. As it mentioned, ERROR3 is completely independent of t in the Durbin's modified method, and is constant. Therefore the Durbin's modified method of numerical inverse Laplace transform is an accurate method (as mentioned in Durbin's paper), and has no error as $t \longrightarrow 0$ (since ERROR3 is independent of time variable). The only limitation that is mentioned in Durbin's paper is that the Durbin's method is valid only for $t \leqslant 2T_t$. On the other hand, ERROR3 is related to T_t by this equation: ERROR3$\leqslant C/[e^{2\varpi T_t} - 1]$, where C is a constant. It seems that ERROR3 increases significantly when $T_t \longrightarrow 0$. But by assuming $\varpi * T_t = 5$–10, T_t will not have any effect on the accuracy. This assumption let us obtain accurate results even if $T_t = 10E - 9$. In this example, although

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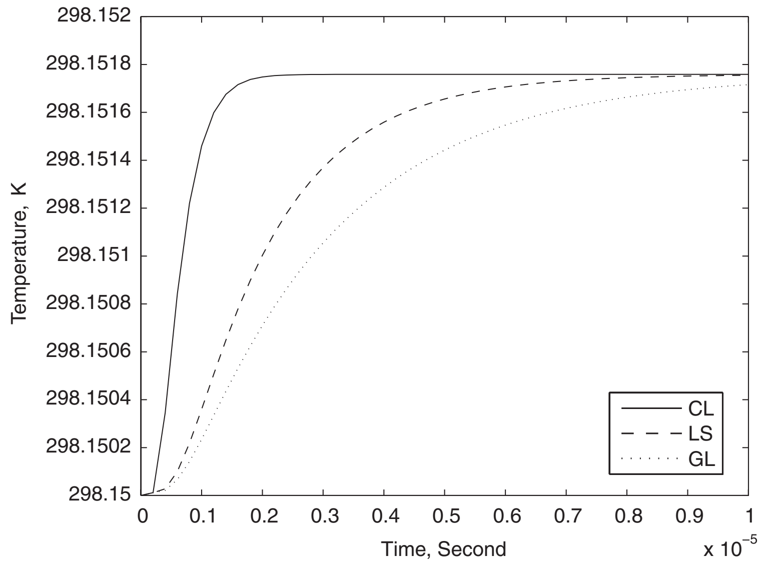

Figure 2. Temperature vs time at the mid-length ceramic-rich side of the functionally graded Ti–6Al–4V/ZrO_2 cylindrical shell for various thermoelasticity models.

the shock period is $1E-6$, T_t is $1E-5$. Substituting $\varpi * T_t = 7.5$ in ERROR3 equation, yields ERROR3$\leqslant C * 3.0590E-7$ (C is a constant with the order of main results). Since the order of radial displacement is $1E-11$, the order of ERROR3 for radial displacement is lower than $1E-18$.

Figure 2 shows the mid-length inner surface temperature vs time for various coupled thermoelasticity theories. It is observed that for the classical coupled thermoelasticity theory 'CL', the temperature suddenly meets maximum. It is due to the parabolic nature of the coupled energy equation, where the metal and ceramic feel the heat at the same time. On the other hand, temperature increasing in the generalized thermoelasticity 'LS' and 'GL' models are less rapidly, due to the hyperbolic type of the coupled energy equation. This is related to the existence of thermal relaxation times. The lateral deflection of the shell middle length vs time is shown in Figure 3. Since 'GL' includes the temperature-rate among the constitutive variables, and does not violate the classical Fourier's law, as 'LS' does, the 'GL' lateral deflection coincides with that of 'CL'. Due to the applied impulsive shock, the shell temperature peaks to a maximum value, and then steadily decrease by time. The peak of temperature affects the axial force, axial moment, and the axial stress, as shown in Figures 4–6. The distribution of axial force vs time for the mid-plane mid-length of the shell is shown in Figure 4. Figure 5 describes the distribution of axial moment vs time for the mid-plane mid-length of the shell. The axial stress vs time for the middle plane mid-length of the shell is plotted in Figure 6. As seen, the axial stress decrease very sharply in 'CL' as excepted. But for two generalized cases 'LS' and 'GL', since the heat conduction equation is hyperbolic-type, the axial stress decrease gently. Since 'LS' is based on the non-Fourier thermal wave model, the 'LS' axial stress does not decrease as gently as 'GL' does. Due to the low order of magnitude of radial displacement, it is observed that the axial force, axial moment and axial

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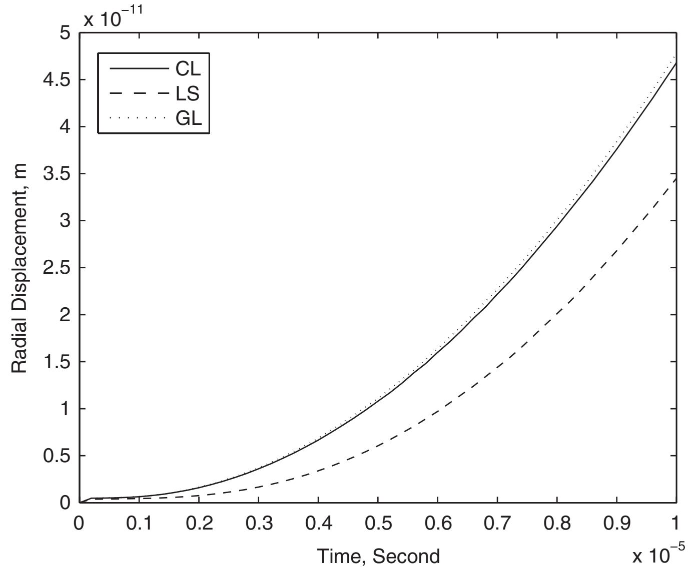

Figure 3. Radial displacement vs time at the middle length of the functionally graded Ti–6Al–4V/ZrO_2 cylindrical shell for various thermoelasticity models.

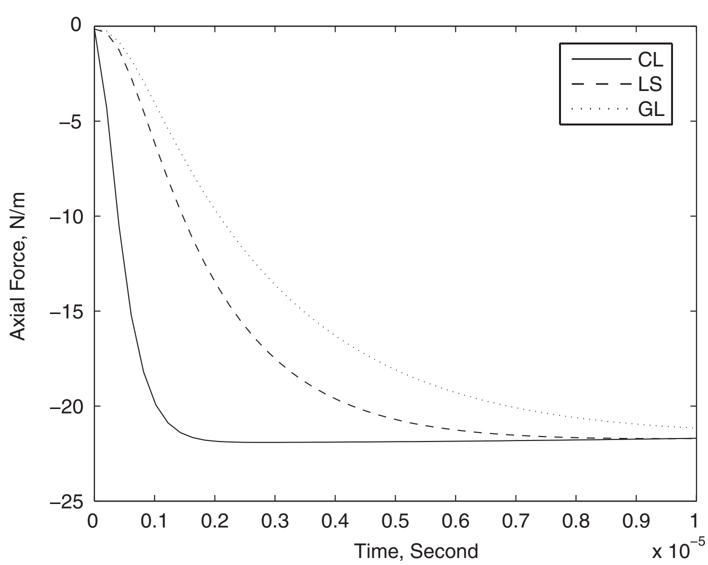

Figure 4. Axial force vs time at the middle length of the functionally graded Ti–6Al–4V/ZrO_2 cylindrical shell for various thermoelasticity models.

stress follow the pattern of temperature. Figures 7 and 8 show the distribution of axial stress vs time for the mid-plane mid-length of the shell for the power law indices of $m_i = 0$ $(i = 1, \ldots, 5)$ (isotropic material) and $m_i = 1$ $(i = 1, \ldots, 5)$ (linear material variation), respectively. Comparing

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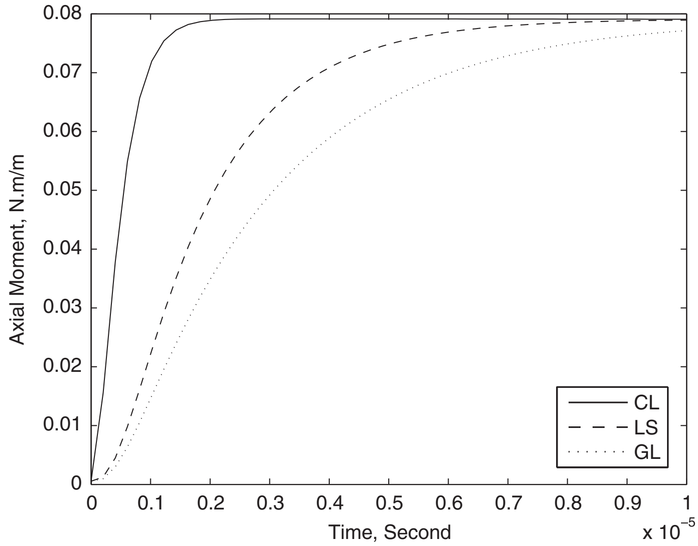

Figure 5. Axial moment vs time at the middle length of the functionally graded Ti–6Al–4V/ZrO_2 cylindrical shell for various thermoelasticity models.

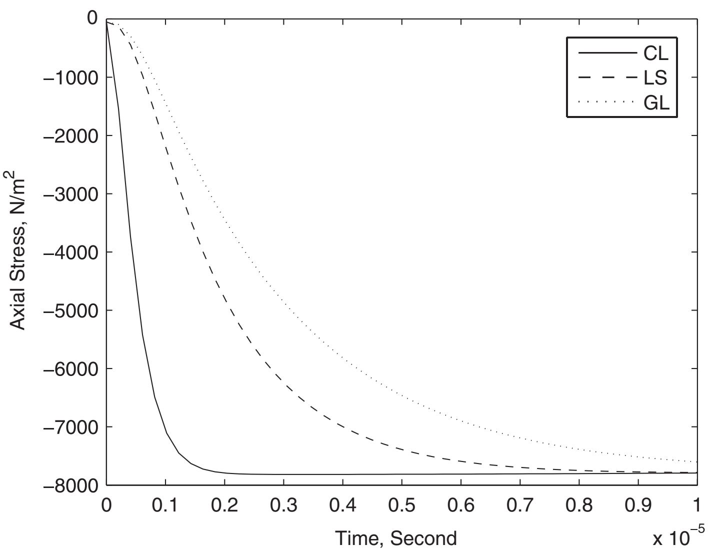

Figure 6. Axial stress vs time at the mid-length ceramic-rich side of the functionally graded Ti–6Al–4V/ZrO_2 cylindrical shell for various thermoelasticity models, where $m_i = 3$ $(i = 1, \ldots, 5)$.

Figures 6–8 illustrate that the axial stress increases as the power law index increases (the shell thickness becomes more ceramic reach).

Now, reconsider the previous example. The length of shell is assumed to be 400 mm, diameter 217 mm, and wall thickness of 2 mm. The ratio R/h is thus 54. Material is the

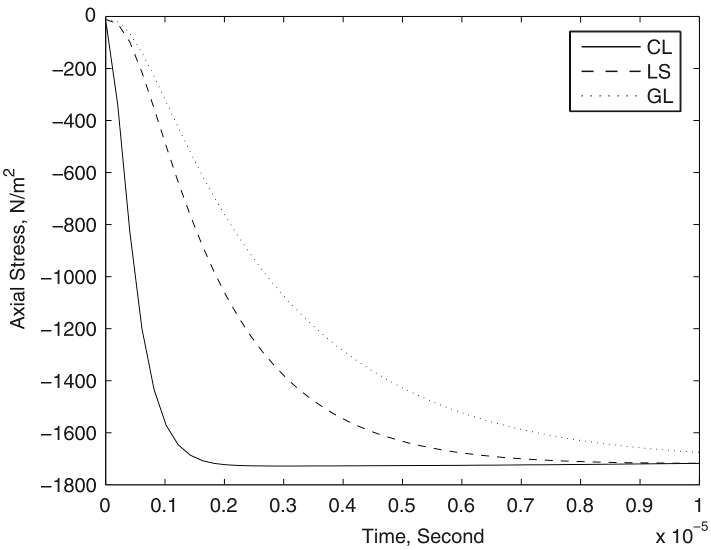

Figure 7. Axial stress vs time at the mid-length ceramic-rich side of the functionally graded Ti–6Al–4V/ZrO_2 cylindrical shell for various thermoelasticity models, where $m_i = 0$ $(i = 1, \ldots, 5)$.

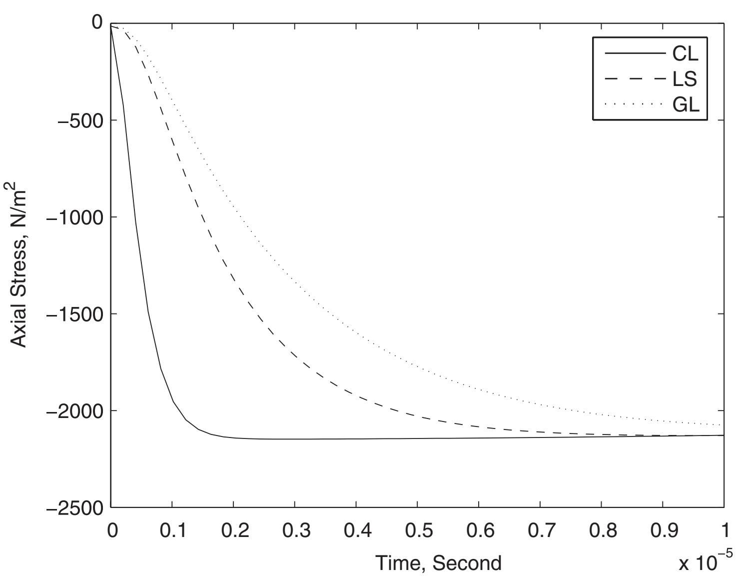

Figure 8. Axial stress vs time at the mid-length ceramic-rich side of the functionally graded Ti–6Al–4V/ZrO_2 cylindrical shell for various thermoelasticity models, where $m_i = 1$ $(i = 1, \ldots, 5)$.

combination of metal and ceramic at the initial temperature 298.15°K and with the material properties shown in Table II. One-half of the cylindrical shell is modelled and is divided into 200 elements of identical lengths along the middle surface. The FGM shell is studied for the power law index of $m_i = 1$ $(i = 1, \ldots, 5)$. Two linear and second-degree temperature gradients

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Table II. Material properties of a special FGM.

Metal	Ceramics
$E = 105.802$ (GPa)	$E = 168.4$ (GPa)
$\nu = 0.2982$	$\nu = 0.2979$
$\alpha = 6.9585 \times 10^{-6}$ (1/K)	$\alpha = 18.4843 \times 10^{-6}$ (1/K)
$\rho = 8.9 \times 10^3$ (kg/m^3)	$\rho = 2.37 \times 10^3$ (kg/m^3)
$k = 6.08$ (W/mK)	$k = 1.775$ (W/mK)
$c = 624.938$ (J/kg K)	$c = 529.027$ (J/kg K)

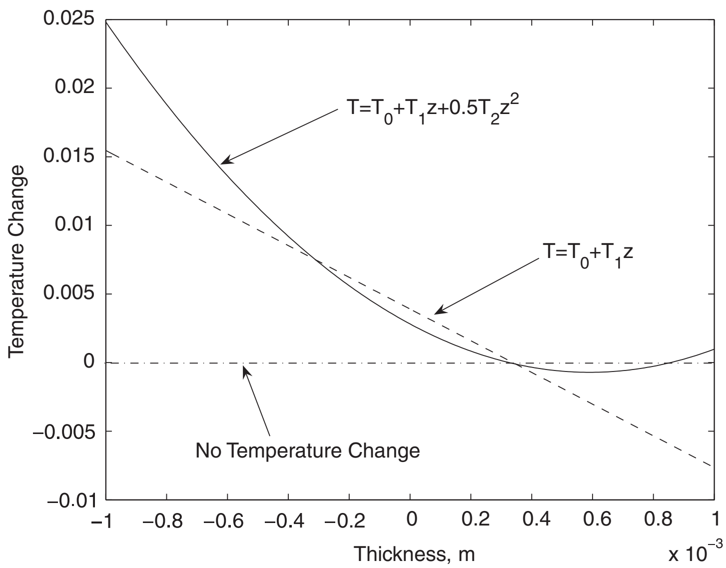

Figure 9. Temperature along the thickness at the mid-length of the functionally graded cylindrical shell with various temperature fields, at $t = 400\,\mu$s.

across the shell thickness, as given by Equations (11) and (12), are assumed and their results are compared at $t = 400\,\mu$s. In this example, the classical coupled thermoelasticity theory is assumed.

Suitable selection of λ's ($\lambda_5 = 0.01$, $\lambda_7 = 1E6$, $\lambda_8 = 1E9$, and other λ's equal 1), specially when temperature gradients across the shell thickness is second-degree, will effect solution duration time and solution convergence. These is because of the one more energy equation that the second-degree temperature problem has, than the linear temperature filed across the shell thickness.

Temperature at the inner surface of the shell is plotted as a function of time for two different temperature distributions across the shell thickness in Figure 9. As shown, the temperature change for linear temperature field is below zero at surface near to the shell outer surface. Due to the physical thermal boundary conditions applied to the shell, negative temperature change is not

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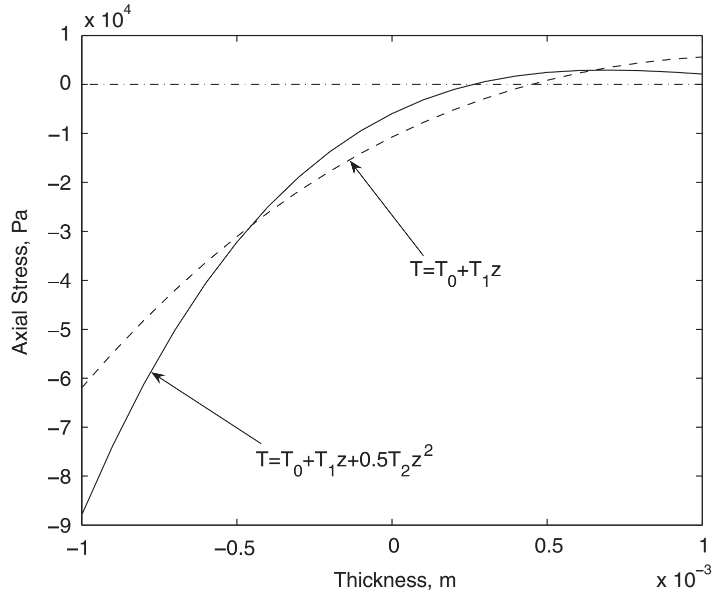

Figure 10. Axial stress distributions along the thickness at the mid-length of the functionally graded cylindrical shell with various temperature fields, at $t = 400\,\mu s$, where $m_i = 1$ $(i = 1, \ldots, 5)$.

satisfied. This error is eliminated by assuming a second-degree temperature field through the shell thickness. Figure 10 shows the axial stress distributions along the shell thickness, with $m_i = 1$ $(i = 1, \ldots, 5)$. As shown, the axial stress of non-linear temperature field is larger than that of linear field. This difference is as the same as the temperature difference, which suggests that the temperature field directly affects the stress field.

For linear temperature field, due to the variation of volume fraction through the thickness, the axial stress distributions across the thickness is non-linear. This result for the isotropic case is linear, as shown in Figure 11. In the case of non-linear temperature field, axial stress vs thickness is highly curved. The axial stress distributions along the shell thickness for the power law index $m_i = 3$ $(i = 1, \ldots, 5)$ is shown in Figure 12.

One may expect to see the stress wave front across the shell thickness, as the lateral shock was applied to the shell. This phenomenon, however, is not observed in the results of this paper (Figures 10–12) and those of Eslami *et al.* [11]. The reason is the selection of the shell model based on the flexural theory and the assumption of temperature field through the thickness. That is, the phenomenon of stress wave front is observed if the theory of elasticity is used [19]. The computed results for the functionally graded shell are compared with that of isotropic shell under sustain thermal shock using the LS model [11]. The method of aforesaid paper is a time marching method, instead of the inverse of Laplace transform method. Also, The results are reduced to the shell with the classical model of thermoelasticity and validated with the known data in the literature [20].

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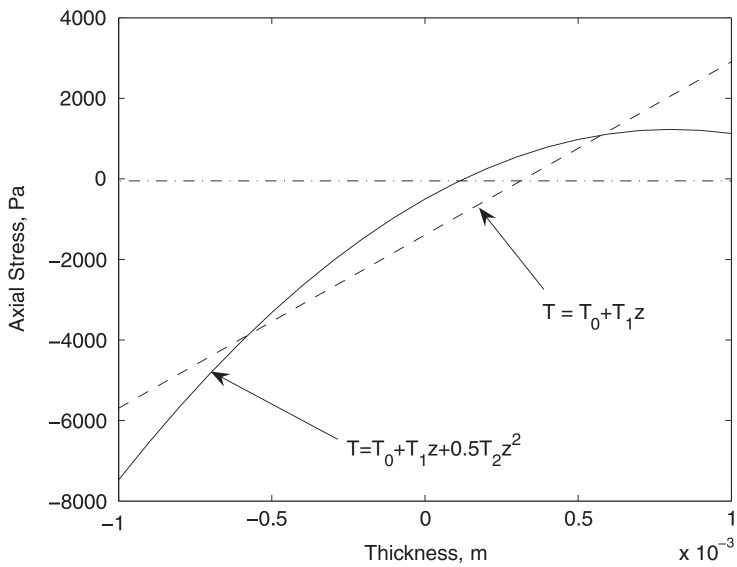

Figure 11. Axial stress distributions along the thickness at the mid-length of the functionally graded cylindrical shell with various temperature fields, at $t = 400\,\mu$s, where $m_i = 0$ $(i = 1, \ldots, 5)$.

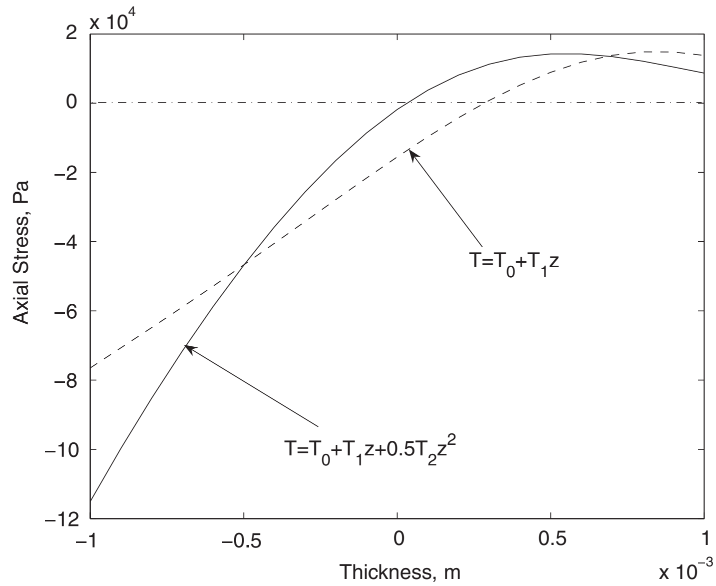

Figure 12. Axial stress distributions along the thickness at the mid-length of the functionally graded cylindrical shell with various temperature fields, at $t = 400\,\mu$s, where $m_i = 3$ $(i = 1, \ldots, 5)$.

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5. CONCLUSIONS

This paper presents the numerical solution for the generalized coupled thermoelasticity of functionally graded cylindrical shells. For the general case, we have provided a numerical solution for the calculation of the displacement, temperature, and thermal stresses. Results for a symmetrically loaded impulsive thermal shock applied to the inner surface are presented for cylindrical shell of functionally graded material with clamped edges.

The method of Laplace transform is used to eliminate the time in the governing finite element equations. Therefore, need for time increment is eliminated in the calculation. The benefit of this method is the selection of short time increments to study the wave propagation, as in the inverse Laplace transform the real magnitude of time increment is arbitrary and may be selected small, as desired.

The flexural theory, as it averages the stress across the shell thickness, is unable to show the wave front. The selection of the shell model based on the elasticity theory, instead of the flexural theory, and dividing the shell into a number of elements through the thickness will lead to appearance of wave front phenomenon.

The generalized thermoelasticity in functionally graded cylindrical shells based on the Lord–Shulman model is compared with the Green–Lindsay model. Since 'GL' does not violate the Fourier thermal wave model, the 'GL' axial stress decrease gently.

A method was introduced to normalize ill-condition matrices in the numerical solutions. Since in shock-type problems finite element matrices become ill-condition, deriving the results is too difficult and sometimes unreachable. Thus, some constants ($\lambda_i,\ i=1,2,\ldots,8$) are declared while changing the governing equations into dimensionless form. Using these constants, the ill-condition global stiffness matrix is normalized. These constants are chosen in such a way not to effect the results of the problem. Since the Lambda values are selected with trial and errors, the solution procedure is only time-consuming, and it depends on the degree of ill-condition of the global stiffness matrix.

APPENDIX A

$$(\Pi_{01}^i, \Pi_{02}^i, \Pi_{03}^i) = \int_z (1, z, z^2)(2z+h)^{(i-1)k}\,\mathrm{d}z, \quad i=1,2,3$$

$$(\Pi_{11}^i, \Pi_{12}^i, \Pi_{13}^i, \Pi_{14}^i) = \int_z (1, z, z^2, z^3)\left(1+\frac{z}{R}\right)(2z+h)^{(i-1)k}\,\mathrm{d}z, \quad i=1,2,3$$

$$(\Pi_{21}^i, \Pi_{22}^i, \Pi_{23}^i) = \int_z (1, z, z^2)\frac{R}{R+z}(2z+h)^{(i-1)k}\,\mathrm{d}z, \quad i=1,2,3$$

$$\Omega_1 = \frac{E_\mathrm{m}}{1-v^2}, \quad \Omega_2 = \frac{E_\mathrm{m}-E_\mathrm{c}}{(1-v^2)(2h)^k}, \quad \Omega_3 = \frac{E_\mathrm{m}\alpha_\mathrm{m}}{1-v}$$

$$\Omega_4 = \frac{E_\mathrm{m}(\alpha_\mathrm{m}-\alpha_\mathrm{c}) + \alpha_\mathrm{m}(E_\mathrm{m}-E_\mathrm{c})}{(1-v)(2h)^k}, \quad \Omega_5 = \frac{(E_\mathrm{m}-E_\mathrm{c})(\alpha_\mathrm{m}-\alpha_\mathrm{c})}{(1-v)(2h)^{2k}}$$

$$\Omega_6 = \frac{E_\mathrm{m}}{2(1+v)}, \quad \Omega_7 = \frac{E_\mathrm{m}-E_\mathrm{c}}{2(1+v)(2h)^k}, \quad \Omega_8 = \rho_\mathrm{m} c_\mathrm{m}$$

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$$\Omega_9 = \frac{\rho_m(c_m - c_c) + c_m(\rho_m - \rho_c)}{(2h)^k}, \quad \Omega_{10} = \frac{(\rho_m - \rho_c)(c_m - c_c)}{(2h)^{2k}}$$

$$\Gamma_{ij}^1 = -\Omega_3 \Pi_{ji}^1 - \Omega_4 \Pi_{ji}^2 - \Omega_5 \Pi_{ji}^3$$

$$\Gamma_{ij}^2 = -\Omega_8 \Pi_{ji}^1 - \Omega_9 \Pi_{ji}^2 - \Omega_{10} \Pi_{ji}^3$$

$$\Gamma_{ij}^3 = k_m \Pi_{ji}^1 + \frac{k_m - k_c}{(2h)^k} \Pi_{ji}^2 \tag{A1}$$

APPENDIX B: NOMENCLATURE

c	specific heat conduction
E	Young's modulus
E_a	average Young's modulus
h	shell thickness
h_{in}, h_{out}	inside, outside convective coefficients
k	thermal conduction coefficient
L	length
$m_1, \ldots, m_5$	volume fraction index
p_{max}	maximum applied load
$p_z(t)$	lateral force
q_i	heat flux
R	mean radius
s	Laplace variable
t	time variable
t_0, t_1, t_2	relaxation times
T	shell temperature
T_0	mean shell temperature
T_1, T_2	gradients through the thickness
T_a	reference temperature
T_∞	ambient temperature
u_0, v_0, w_0	displacement components
u_1, u_2, u_3	displacement components
x, θ, z	shell coordinates

Greek letters

α	thermal expansion coefficient
α_a	average thermal expansion coefficient
β_{ij}	general anisotropic coupling coefficient tensor
ΔT	temperature change
λ_i	normalization constants
λ_a, μ_a	average Lamé constants
ν	Poisson's ratio

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ρ	mass density
ρ_{a}	average mass density
ψ_x, ψ_θ	middle plane tangents rotations

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